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UREA CYCLE

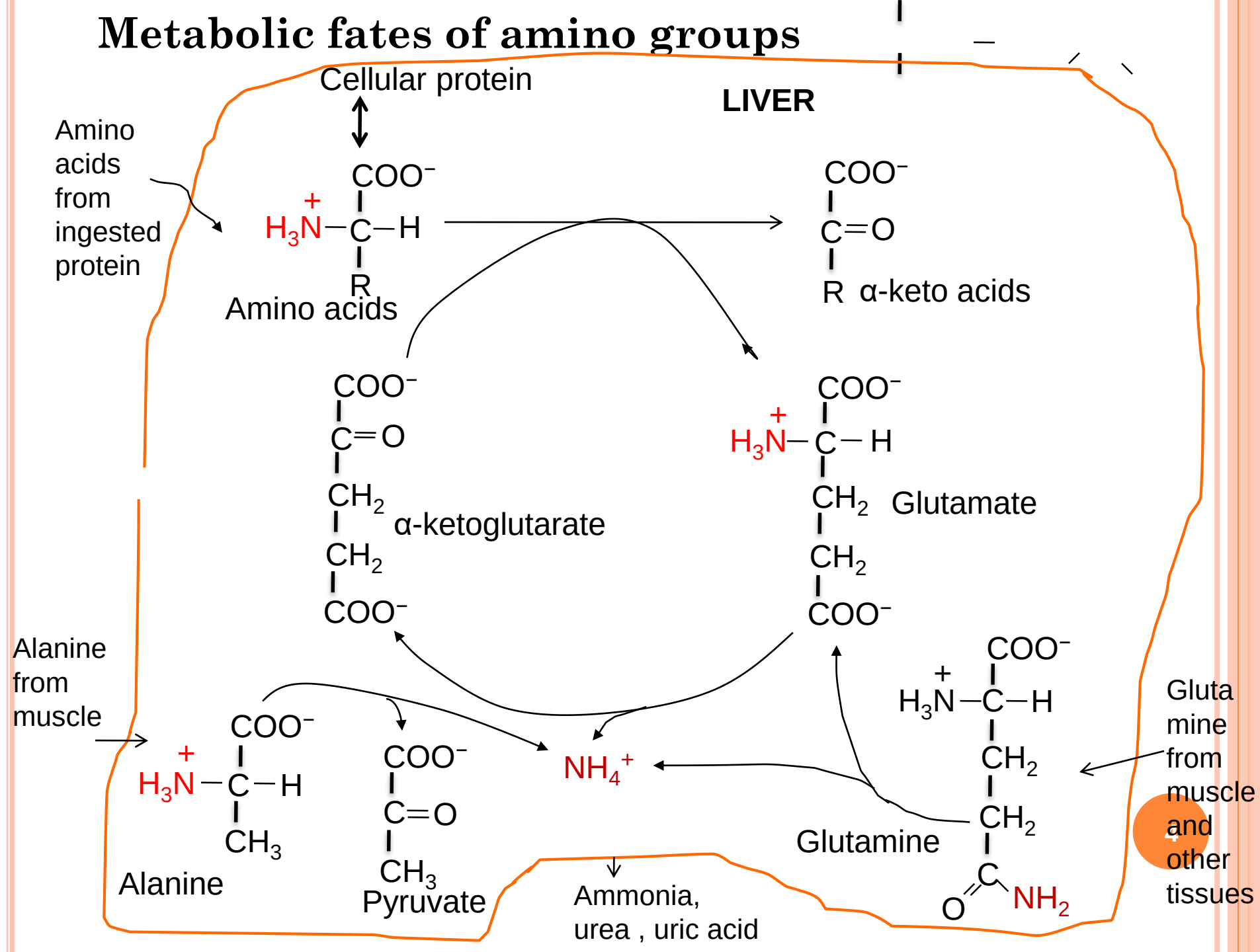
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AMINO ACID OXIDATION AND PRODUCTION OF UREA

- In animals amino acids undergo oxidative degradation in three different metabolic circumstances
 - During normal synthesis and degradation of cellular proteins , some amino acids released undergo oxidative degradation if they are not needed for new protein synthesis
 - When a diet is rich in protein and the ingested amino acids exceeds the body's needs for protein synthesis, the surplus is catabolized; amino acids can not be stored
 - During starvation or in diabetes mellitus when carbohydrates are either unavailable or not properly utilized, cellular proteins are used as fuel
- In all cases amino acids lose their amino groups to form α -keto acids, the 'carbon skeletons' of amino acids
 - α -keto acids
 - Undergo oxidation to CO_2 and H_2O
 - More frequently provide three- or four-carbon units that can be converted by gluconeogenesis into glucose

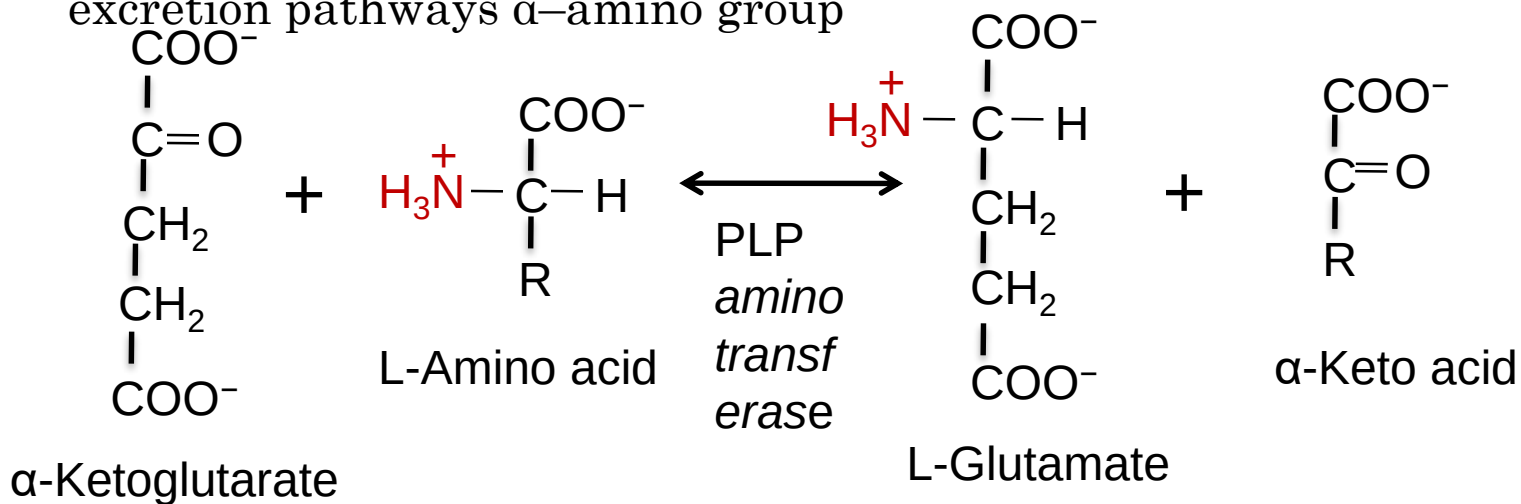
- Amino acids derived from dietary protein are the source of most amino groups
- Most amino acids are metabolized in the liver
 - Ammonia is generated in the process
 - Some of the ammonia is recycled and used in a variety of biosynthetic process
 - Excess ammonia is excreted
 - Directly as ammonia (In ammonotelic animals as most aquatic vertebrates such as bony fish)
 - After being converted to urea (In ureotelic animals such as many terrestrial vertebrates)
 - After being converted to uric acid (in uricotelic animals such as birds and reptiles)
- Excess ammonia generated in extrahepatic tissues travels to the liver (in the form of amino groups) for conversion to the form in which its excreted
- Glutamate and glutamine play critical roles in nitrogen metabolism
 - In the cytosol of hepatocytes amino groups from most amino acids are transferred to α -ketoglutarate to form glutamate
 - Glutamate is then transported into the mitochondria where the amino group is removed to form NH_4^+
 - Excess ammonia generated in most other tissues is converted to the amide nitrogen of glutamine, which passes to the liver, then into the mitochondria of hepatocytes
 - In muscle, excess amino groups are generally transferred to pyruvate to form alanine which transports amino groups to the liver

Metabolic fates of amino groups



Pyridoxal phosphate participates in the transfer of α -amino groups to α -ketoglutarate

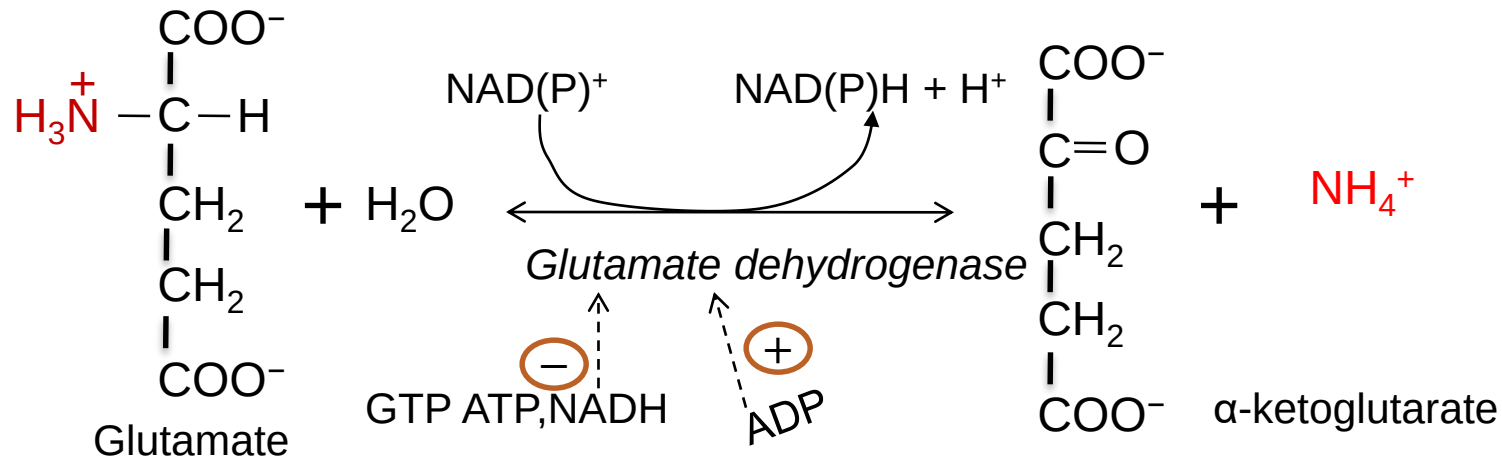
- The first step in the catabolism of most L-amino acids is the removal of the α -amino groups catalyzed by aminotransferases or transaminases
- The α -amino group is transferred to the α -carbon atom of α -ketoglutarate, leaving behind the corresponding α -keto acid analog of the amino acid
- Transamination results in amino groups from many different amino acids being collected in the form of glutamate.
 - This is important because L-glutamate is the only amino acid that can undergo oxidative deamination at an appreciable rate
- Glutamate functions as an amino group donor for biosynthetic pathways or excretion pathways α -amino group



- Most but not all amino acids are substrates for transamination
 - Exceptions include; lysine, threonine, proline and hydroxyproline

- There are a number of different aminotransferases
 - Many are specific for α -ketoglutarate as the amino group acceptor but differ in their specificity for the L-amino acids
 - Enzymes are named for the amino group donor (eg alanine aminotransferase, aspartate aminotransferase)
 - Aminotransferases have pyridoxal phosphate (PLP) as the prosthetic group
 - PLP functions as an intermediate carrier of amino groups at the active site of aminotransaminases
- Measurements of alanine aminotransferase and aspartate aminotransferase in blood serum is important in medical diagnosis
 - Alanine aminotransferase (ALT; also called glutamate-pyruvate transaminase, GPT) and aspartate aminotransaminase (AST; also called glutamate-oxaloacetate transaminase, GOT) are important in the diagnosis of heart and liver damage caused by heart attack, drug toxicity or infection
 - After a heart attack SGPT, SGOT (S for serum) and serum creatine kinase can provide information about severity of the damage
 - Creatine kinase is the first enzyme to appear after a heart attack; it also disappears quickly from the plasma
 - GOT is the next to appear and GPT follows later
 - Lactate dehydrogenase also leaks from injured heart muscle
 - SGOT and SGPT tests are important to determine whether people exposed to carbon tetrachloride, chloroform etc have suffered liver damage. Various enzymes may leak due to liver damage but aminotransferases are most useful for monitoring because they are very active in the liver

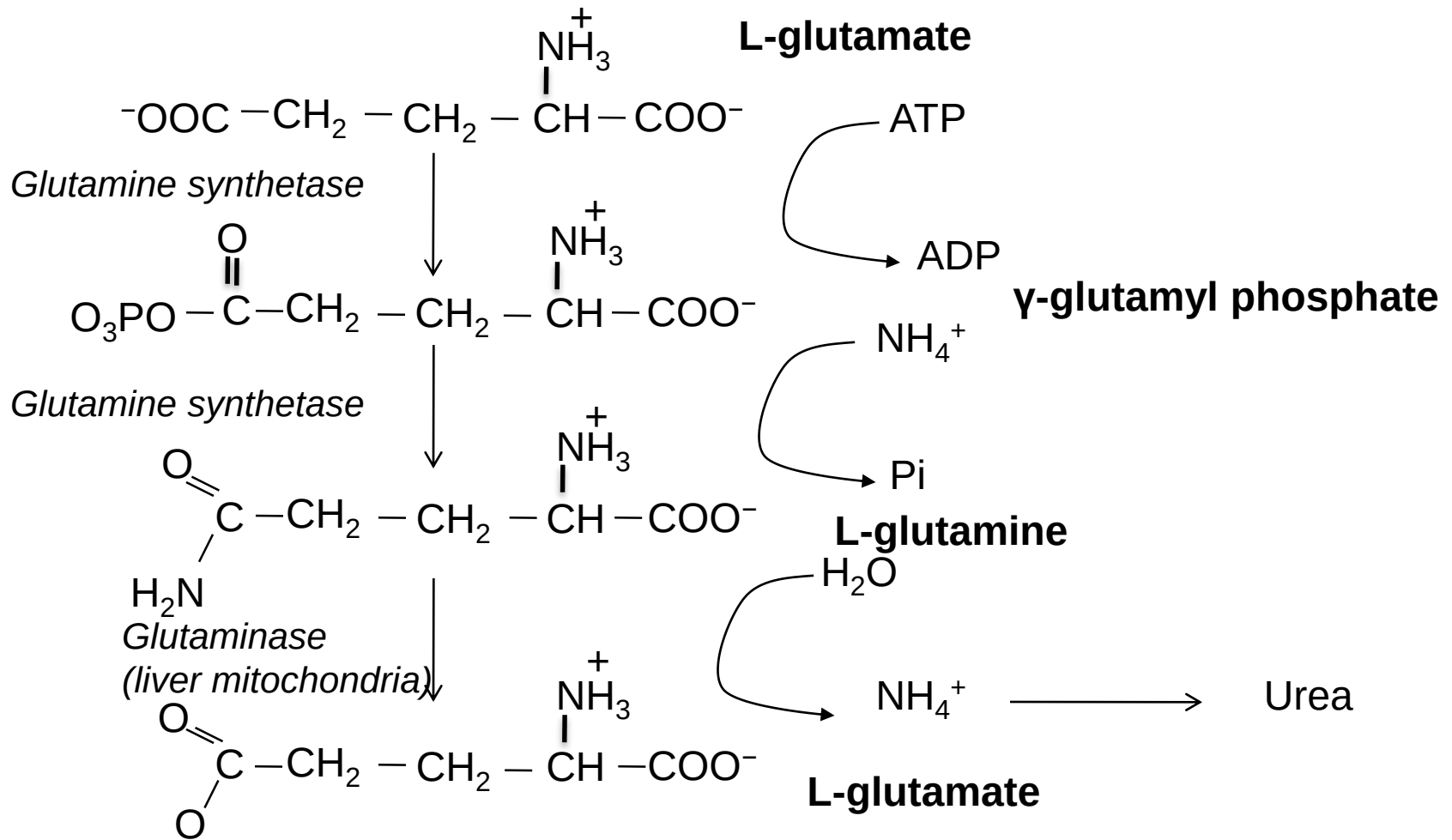
Glutamate releases ammonia in the liver



- In hepatocytes, glutamate is transported from the cytosol into the mitochondria where it undergoes oxidative deamination catalyzed by L-glutamate dehydrogenase
- The enzyme can use either NAD^+ or NADP^+ as acceptor of reducing equivalents
- The combined action of an aminotransferase and glutamate dehydrogenase is referred to as transdeamination
- The α -ketoglutarate produced by deamination of glutamate can be used in the citric acid cycle and for gluconeogenesis

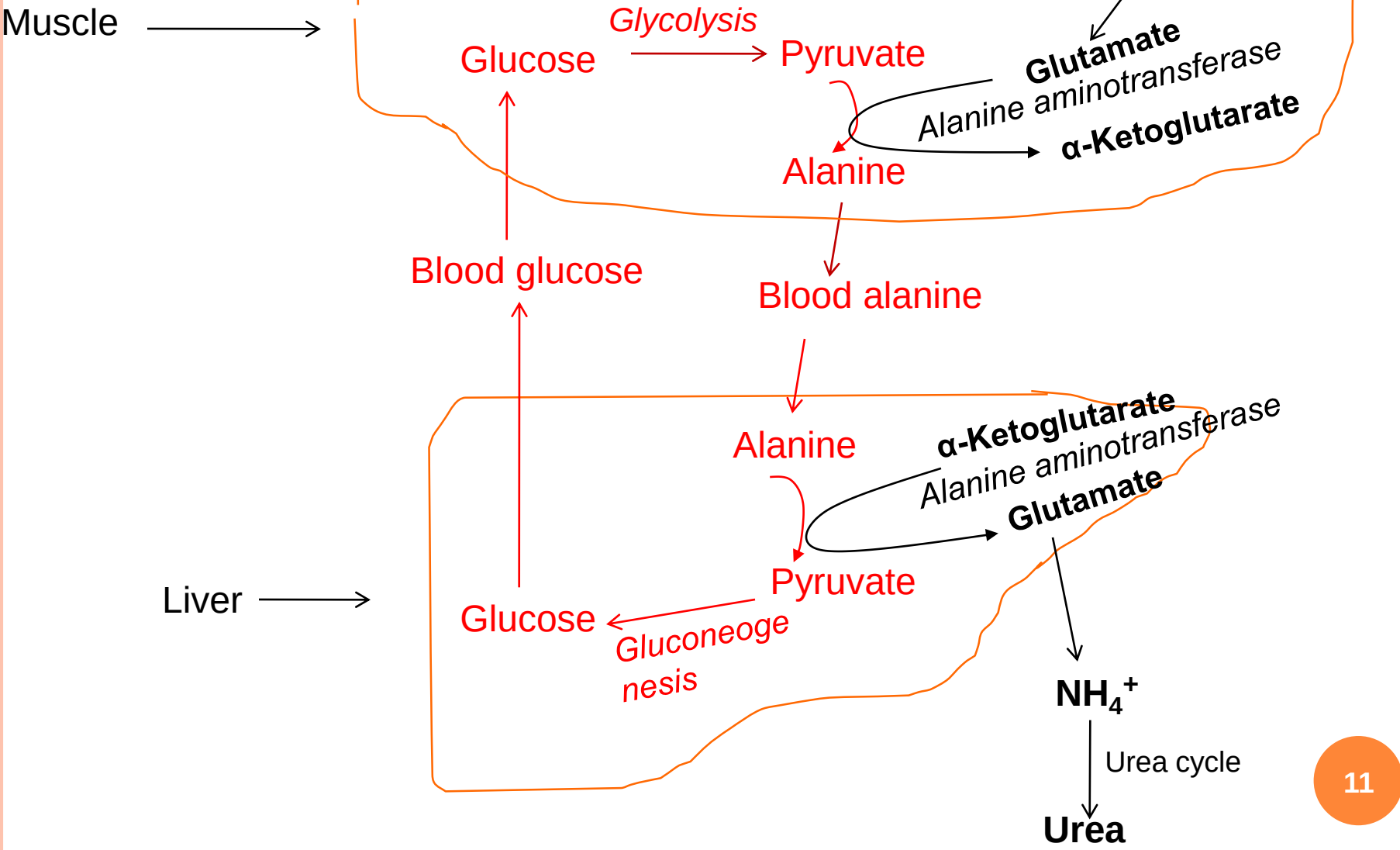
Glutamine transports ammonia in the blood stream

- Ammonia is toxic to animal tissues
- Symptoms of ammonia intoxication include a flapping tremor, slurring of speech, blurring of vision and in severe cases coma and death
- Excess ammonia is converted to a nontoxic compound before export from extrahepatic tissues to the liver and kidneys via the blood stream
- L-glutamine plays a role in transport of ammonia from extrahepatic tissues
- In extrahepatic tissues free ammonia generated is combined with glutamate to form glutamine by the action of glutamine synthetase
 - First glutamate and ATP react to form ADP and a γ -glutamyl phosphate intermediate that reacts with ammonia to produce glutamine and inorganic phosphate
 - Thus glutamine is
 - A nontoxic transport form of ammonia
 - A source of amino groups in a variety of biosynthetic reactions
- Glutamine in excess of that required for biosynthesis is transported in the blood to the liver and kidneys for processing
 - The enzyme glutaminase converts glutamine to glutamate and NH_4^+ . Provides another source of ammonia to be disposed of by urea synthesis
 - Glutamate is further processed in the liver by glutamate dehydrogenase, releasing more ammonia and producing carbon skeletons for metabolic fuel



Alanine transports ammonia from muscles to the liver

- Alanine transports amino groups in a nontoxic form via a glucose-alanine cycle
- In muscle amino groups are collected in the form of glutamate by transamination
- Glutamate can
 - Be converted to glutamine for transport to the liver
 - Transfer its amino group to pyruvate forming alanine . Catalyzed by alanine amino transferase
 - Alanine passes into the blood stream and travels to the liver
 - In the cytosol alanine amino transferase transfers the amino group from alanine to α -ketoglutarate forming pyruvate and glutamate
 - Glutamate then enters the mitochondria where the glutamate dehydrogenase reaction releases NH_4^+ or can undergo transamination reaction with oxaloacetate to form aspartate, another nitrogen donor in the urea cycle



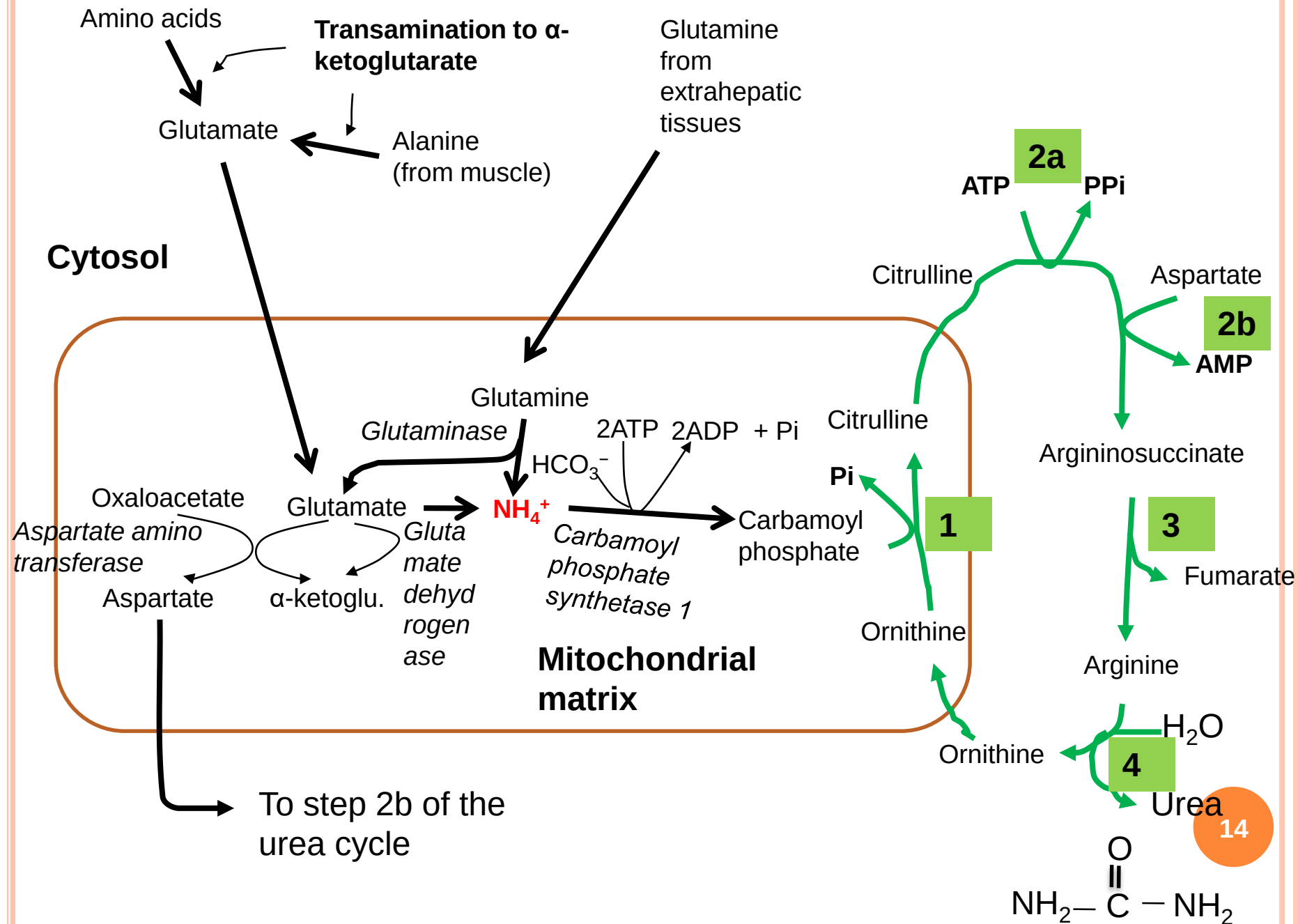
Nitrogen excretion and the urea cycle

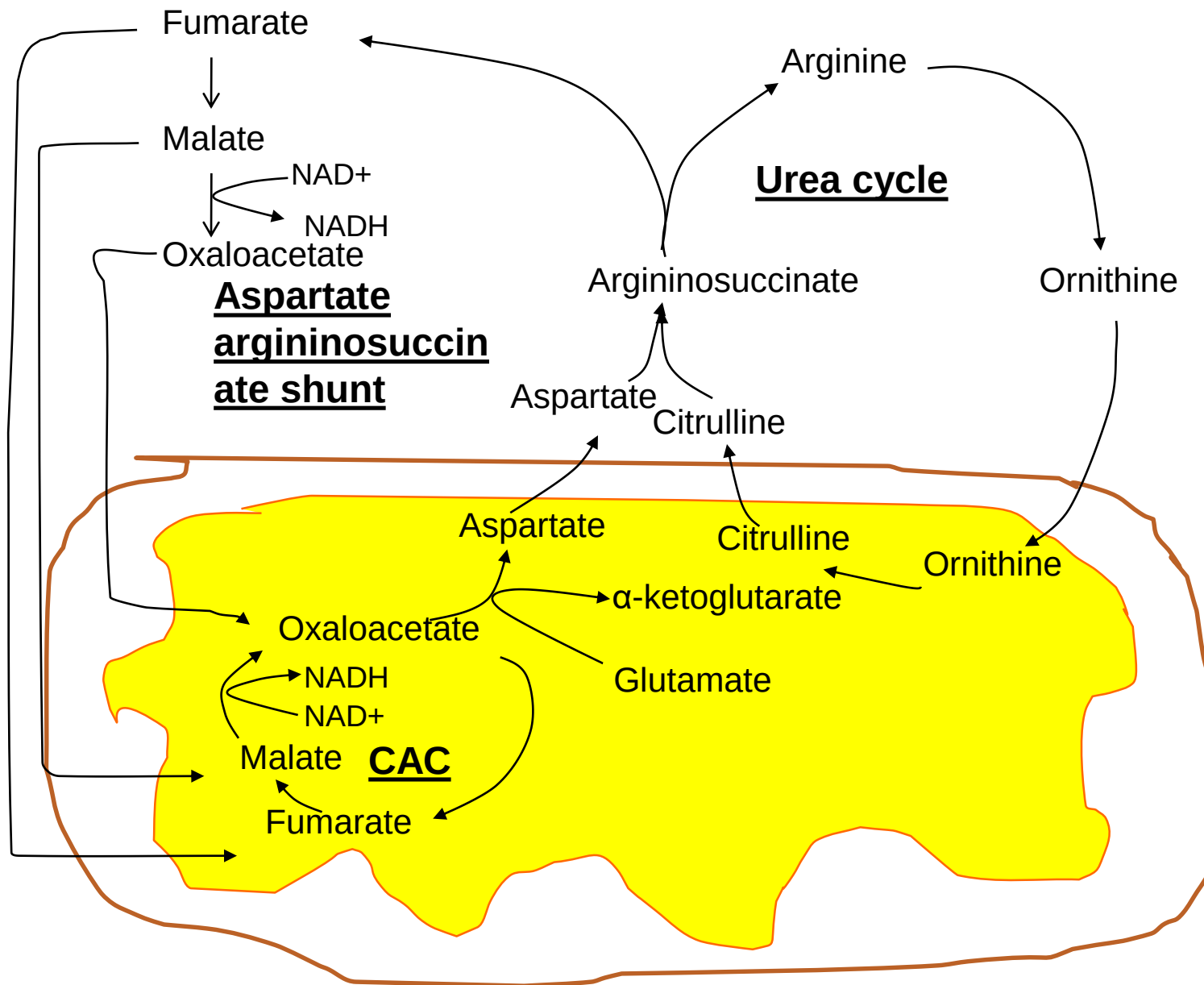
- In ureotelic organisms, the ammonia deposited in the mitochondria of the hepatocytes is converted to urea in the urea cycle
- Urea production occurs mainly in the liver and is the fate of most of the ammonia channeled there
- Urea passes into the blood stream to the kidneys where its excreted

Urea is produced from ammonia in five enzymatic steps

- Source of the first amino group to enter the cycle
 - Ammonia from glutamate and glutamine
 - From the intestine via the portal vein where its produced by bacterial oxidation of amino acids
- Ammonia together with CO_2 (as HCO_3^-) produced by the mitochondrial respiration form carbamoyl phosphate in the mitochondrial matrix
 - ATP- dependent reaction catalyzed by carbamoyl phosphate synthetase I, a regulatory enzyme
- Carbamoyl phosphate (an activated carbamoyl group donor) enters the cycle which has four enzymatic steps
 - Carbamoyl phosphate donates its carbamoyl group to ornithine to form citrulline, with release of Pi (Step 1)
 - Catalyzed by ornithine transcarbamoylase. Citrulline that results passes from

- The second amino group is introduced by aspartate by a condensation reaction between the amino group of aspartate and the ureido group of citrulline, forming argininosuccinate (Step 2)
 - Reaction is catalyzed by argininosuccinate synthetase. Requires ATP and passes through a citrullyl-ATP intermediate
- The argininosuccinate is then reversibly cleaved by argininosuccinate lyase to form free arginine and fumarate (Step 3)
 - Fumarate enters the mitochondria to join a pool of citric acid cycle intermediates
- Arginase then splits the arginine to yield urea and ornithine
 - Ornithine is transported into the mitochondria to initiate another round of the urea cycle
- Aspartate formed in the mitochondria by transamination between oxaloacetate and glutamate can be transported into the cytosol, where it serves as nitrogen donor in the urea cycle reaction catalyzed by argininosuccinate synthetase
- The urea cycle and the citric acid cycle can be linked via fumarate
 - The cycle linking these two cycles is the aspartate-argininosuccinate shunt





The activity of the urea cycle is regulated at two levels

- The flux of nitrogen through the urea cycle varies with the organism's diet
 - On high protein diet , the carbon skeletons of amino acids are used for fuel producing much urea from excess amino groups
 - During prolonged starvation, breakdown of muscle protein supplies much of the metabolic energy and this results in increased urea production
 - All five enzymes involved in urea production are synthesized at higher rates in starving animals and in animals on high-protein diet
- Allosteric regulation of carbamoyl phosphate synthetase I
 - Activated by N-acetylglutamate

Pathway inter-connections reduce the energetic cost of urea synthesis

- The synthesis of one molecule of urea requires four high-energy phosphate groups
 - Two are required to make carbamoyl phosphate
 - A second required to make argininosuccinate undergoes the pyrophosphate cleavage to AMP and PPi which is hydrolyzed to 2Pi
 - Urea cycle also causes a net conversion of oxaloacetate to fumarate (via aspartate), and the regeneration of oxaloacetate produces NADH in the malate dehydrogenase reaction. Each NADH can generate up to 2.5 ATPs reducing the overall energetic cost of urea synthesis